Andrew McGallian

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Geospatial developer who ships production tools: a published QGIS plugin with its own auto-update repository, and a containerized Flask application serving 150 years of Chicago environmental data. MS Environmental Science, University of Chicago.

EDUCATION

University of Chicago Chicago, IL
Master of Science, Environmental Science | GPA 3.86 Jun 2026

University of Chicago Chicago, IL
Bachelor of Science, Environmental Science; Minor in Geographic Information Science Jun 2025

MS Capstone: evaluated the efficacy of cloud seeding through literature synthesis and quantitative reanalysis of historical seeding studies.

PROJECTS

TKAP Tools — QGIS Plugin — *Python, PyQGIS, GPL-2.0 | github.com/amcgallian/tkap-qgis*

- Built and published a QGIS plugin used by an international archaeological field team, distributed via a self-hosted plugin repository so QGIS delivers updates automatically.
- Shipped three tools: GNSS points to stratigraphic-unit polygons, phase splitting with plan export, and section drawing over rectified wall photography.
- Wrote the project's field documentation and set up the release process so the team installs and updates the plugin through QGIS itself.

Chicago Environmental Atlas — *Flask, Docker, Leaflet, Vega-Lite | github.com/amcgallian*

- Built and deployed a containerized web app mapping 150 years of Chicago environmental change (1870–2025), with a timeline driving Landsat rasters and vector layers together.
- Assembled and served roughly 10 GB of processed Landsat tiles and vector layers, including multi-gigabyte biodiversity datasets, behind a Flask API.

EXPERIENCE

Geomatics Research Assistant Konya, Türkiye
Türkmen-Karahöyük Archaeological Project, Institute for the Study of Ancient Cultures Jun 2026 – Aug 2026

- Built the field team's QGIS plugin (above), cutting SU polygon drawing from hours to tens of minutes by automating point-to-polygon generation from Emlid GNSS exports.
- Digitized 1,000+ new stratigraphic units and flew ~20 drone surveys producing orthomosaics and DEMs across a 35-hectare site, 200+ report-quality maps and phase plans.

Data Visualization Fellow Chicago, IL
University of Chicago Library, Center for Digital Scholarship Sep 2025 – Jun 2026

- Built and deployed the Chicago Environmental Atlas and the dissertation topic analysis (above) as the fellowship's two project tracks, from data acquisition through production deployment.
- Acquired and processed archival, satellite, and public environmental datasets, georeferencing historical city boundaries by decade from 1870 and assembling Landsat coverage across five decades.

Digital Scholarship & GIS Assistant Chicago, IL
University of Chicago Library Jun 2025 – Jun 2026

- Inventoried and standardized several thousand records in the university's ArcGIS Online catalog via Python workflows, replacing a manual audit process.
- Designed and taught workshops on ArcGIS Pro, ArcGIS Online, geocoding, and web mapping, building the instructional datasets and technical documentation.

Undergraduate Research Assistant Chicago, IL
Environmental Frontiers, Mansueto Institute for Urban Innovation Jun 2024 – Jul 2025

- Built an iOS app (SwiftUI, CoreLocation) for GPS-tagged field logging and operated M-TEMP sensor carts across Chicago (Jun–Sept 2024), then processed the data against Landsat TIRS.

TECHNICAL SKILLS

Languages & Tools: Python (rasterio, GeoPandas, Pandas, NumPy), SQL, JavaScript (ES6), R, Rust, Git/GitHub, Docker, Flask, PostGIS, Jupyter

GIS & Remote Sensing: ArcGIS Pro, ArcGIS Online, Field Maps, StoryMaps, QGIS (plugin development), Leaflet, geodatabases, cartography, spatial analysis, remote sensing, photogrammetry

3D & Visualization: Blender, Vega-Lite, Matplotlib

Field & Survey: GNSS/RTK (Emlid), drone survey, orthomosaic production, environmental monitoring and sampling, sensor deployment